LMXB Formation Rate in the Nucleus of M31

Eric Pfahl — KITP — 04/20/06

Consider a particular single star or binary in the nucleus of M31. This target object has a linear size s, which is either the stellar radius if the object is single or the semimajor axis if the object is a binary. In order for the target to capture a NS, the pair must pass within a mutual separation of $\lesssim s$. We associate with the capture process a cross section $\Sigma(u)$, where u is the relative speed at infinity between the center of mass of the object and the NS. It is convenient to cast the cross section as

$$\Sigma(u) = Au^{-2}g(u) \ , \tag{1}$$

where A is a constant, and $g(u)$ is a dimensionless function. For the processes being considered, A is of order $\pi G M_t s$, where M_t is the total mass of the interacting system of stars. Let us assume that the density profiles of the targets and the NSs are spherically symmetric, and denote these densities by $n_t(r)$ and $n_{\rm ns}(r)$. As an additional simplification, we assume that the distributions of speeds for the objects and NSs are both Maxwellian with the same one-dimensional velocity dispersion σ. (The actual situation is a bit tricky, since the Keplerian potential of the $\sim 10^8\,M_\odot$ central black is starting the dominate within $\simeq$10–20 pc.) It follows that the distribution of relative speeds is likewise Maxwellian:

$$f(u) = \sqrt{\frac{2}{\pi}}\frac{u^2}{\bar{\sigma}^3}e^{-u^2/2\bar{\sigma}^2} \ , \tag{2}$$

where $\bar{\sigma} = \sqrt{2}\sigma$. Hereafter, we assume that σ is independent of r in the region of interest. For $10 < r < 100\,{\rm pc}$ in the center of M31, σ varies little and has a characteristic value near $150\,{\rm km\,s^{-1}}$. We can now write the probability per unit time that a certain object captures a NS at position r:

$$\gamma(r) = n_{\rm ns}(r)\int_0^\infty du f(u) u \Sigma = \frac{2}{\sqrt{\pi}}\frac{n_{\rm ns}(r)A}{\sigma}\int_0^\infty dz z e^{-z^2}g(z) \ , \tag{3}$$

where $z = u/\sqrt{2\bar{\sigma}^2}$. Note that $n_{\rm ns}A/\sigma$ carries the units of a rate. Letting I represent the undetermined z integral and $A = \pi G M_t s$, we find that

$$\gamma(r) \sim 10^{-15}\left(\frac{n_{\rm ns}}{10^3\,{\rm pc^{-3}}}\right)\left(\frac{M_t}{M_\odot}\frac{s}{R_\odot}\frac{150\,{\rm km\,s^{-1}}}{\sigma}\right) I \ {\rm yr^{-1}} \ . \tag{4}$$

The chosen scale value for the density in eq. (4) is representative of the total stellar density at $r \sim 10\,{\rm pc}$ from the center of M31 (see the Appendix); the density of NSs is probably $\sim$1% of this value. Even though the capture rate per binary is very small, the total rate summed over all binaries may be quite substantial.

Now, integrate the product $\gamma(r)n_t(r)$ over a spherical shell with inner and outer radii $r_{\rm min}$ and $r_{\rm max}$ to obtain the total NS capture rate in the region:

$$\Gamma(r_{\rm min}, r_{\rm max}) = 8\sqrt{\pi}\frac{A}{\sigma}I\int_{r_{\rm min}}^{r_{\rm max}} dr r^2 n_t(r) n_{\rm ns}(r) \ . \tag{5}$$

Let $n_t = \beta_t n$ and $n_{\rm ns} = \beta_{\rm ns} n$, where n is the total number density of stars, and, as a concrete example, suppose $n(r) = n_0(r/r_0)^{-2}$. We then have

$$\Gamma(r_{\rm min}, r_{\rm max}) = 8\sqrt{\pi}\beta_t\beta_{\rm ns}r_0^3 n_0^2\frac{A}{\sigma}(\xi_{\rm min}^{-1} - \xi_{\rm max}^{-1})\, I \ , \tag{6}$$

where $\xi = r/r_0$. Numerically, we find

$$\Gamma(r_{\rm min}, r_{\rm max}) \sim 10^{-8}\beta_t\beta_{\rm ns}\left(\frac{r_0}{10\,{\rm pc}}\right)^3\left(\frac{n_0}{10^3\,{\rm pc^{-3}}}\right)^2\left(\frac{M_t}{M_\odot}\frac{s}{R_\odot}\frac{150\,{\rm km\,s^{-1}}}{\sigma}\right)(\xi_{\rm min}^{-1} - \xi_{\rm max}^{-1})\, I \ , \tag{7}$$

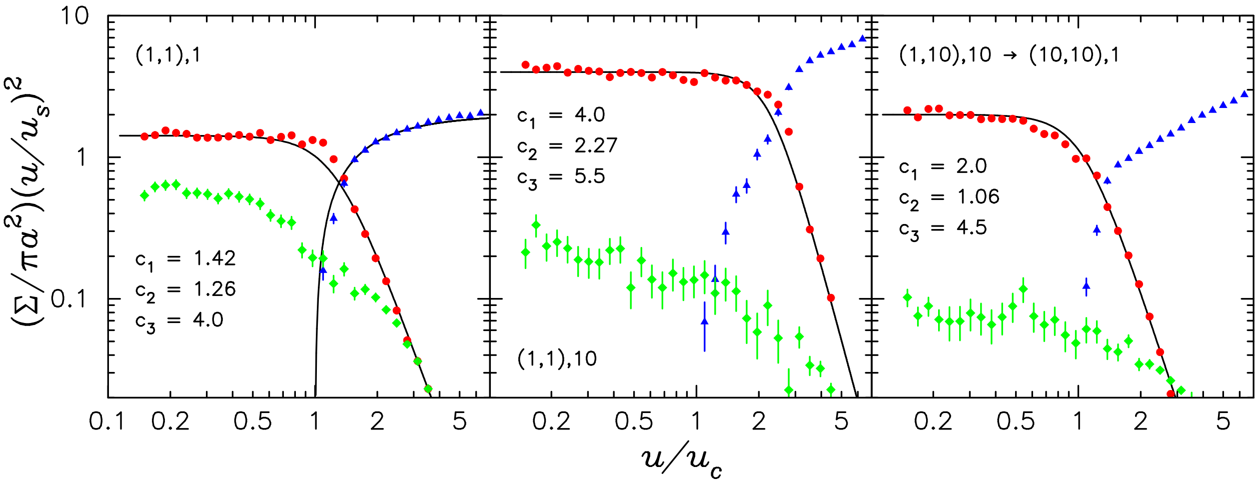

Figure 1: Exchange (red circles), ionization (blue triangles), and collision (green diamonds) cross sections for three different mass groups (cross sections and fits computed by EP). The scale speeds are $u_s = \sqrt{GM_t/a}$ and $u_c = \sqrt{GM_1M_2M_t/(M_bM_3a)}$. The stellar masses are given by labels such as $[(1,1),10]$, which symbolizes that the incident binary (in parentheses) has two stars of 1 mass unit each, while the incoming single star has a mass of 10 units.

where we again assumed $A = \pi GM_t s$. Once we know the cross section, the quantities A and I are determined.

There are three relevant processes—not restricted to NS capture—for which an analytic cross section is readily obtained. These are

Direct Collision:

$$\Sigma_c = (2\pi sGM_t)u^{-2}\left[1 + \left(\frac{4\sigma^2 s}{GM_t}\right)z^2\right], \tag{8}$$

Exchange:

$$\Sigma_e = (c_1\pi sGM_t)u^{-2}\left[1 + \left(\frac{2\sigma}{c_2u_c}\right)^{c_3}z^{c_3}\right]^{-1}, \tag{9}$$

Ionization:

$$\Sigma_i = (2\pi sGM_t)u^{-2}\left[1 - \left(\frac{2\sigma}{u_c}\right)^{-3/2}z^{-3/2}\right], \quad (u > u_c)\ . \tag{10}$$

In each case, A and $g(z)$ are apparent. Note that in the collision cross section, s represents some multiple of the stellar radius, while for exchange and ionization, s is the binary semimajor axis. The exchange and ionization cross sections are smooth fitting functions over the full range of incident speeds (see Fig. 1). In fact, the form of the ionization cross section in eq. (10) is valid only when all the masses are equal. The critical speed at infinity u_c separates positive and negative total system energies; ionization is only possible when $u > u_c$. Hereafter, we assume, without much loss in generality, that all stars have the same mass M, so that $u_c = \sqrt{3GM/2s}$; the parameters c_i are listed in the left panel of Fig. 1.

In the case of direct collisions (or tidal capture), the z integral in eq. (3) is analytic, with the result

$$I_c = \frac{1}{2}\left[1 + \left(\frac{2\sigma^2 s}{GM}\right)\right]\ . \tag{11}$$

When $\sigma = 150\,\mathrm{km\,s^{-1}}$, $M = 1\,M_\odot$, and $s = 1\,R_\odot$, we see that the second quantity in brackets is $\simeq$0.24. If s is increased to $\gtrsim 4\,R_\odot$, as in the case of a subgiant or giant target star, the geometric cross section begins to dominate. Nonetheless, with σ constant, the total collision rate per unit volume still scales with radius as $n^2(r)$ (see eq. [5]) as long as β_t and $\beta_{\rm ns}$ are constant.

For exchange and ionization, we appealed to numerical integration to evaluate the speed integrals, again assuming that $\sigma = 150\,{\rm km\,s^{-1}}$, and that all masses equal $M = 1\,M_\odot$. The values of the z integrals I_e and I_i are shown in the Table below for several values of s.

s(AU)	I_e	I_i
0.1	0.129	0.286
0.3	0.052	0.384
0.5	0.033	0.415
1.0	0.017	0.445

It is noteworthy that I_e varies nearly as s^{-1}, while we might expect $I_e \propto s^{-2}$, as in the high-speed limit of the cross section in eq. (9). The reason for the slower trend is that the high-speed piece of $g(z)$ ($\propto z^{-4}$) effectively truncates the integrand at $z \simeq u_c/2\sigma$, and since $(u_c/2\sigma)^2 \ll 1$ for interesting pre-exchange binaries, the largest contribution to the integral is

$$I_e \sim \int_0^{u_c/2\sigma} dzz = \frac{3GM}{16\sigma^2 s} \sim 0.1\left(\frac{M}{M_\odot}\frac{0.1\,{\rm AU}}{s}\right)\left(\frac{150\,{\rm km\,s^{-1}}}{\sigma}\right)^2 . \tag{12}$$

This proportionality, along with $A \propto s$, means that the exchange rate is essentially independent of s. All this being said, it is still true that $I_i \gg I_e$ for $s \gtrsim 0.3\,$AU, and so a NS is much more likely to ionize such a binary than it is to be captured. However, this is not as bad as it might seem. A binary with $s \lesssim 1\,$AU at $r \gtrsim 10\,$pc in the nucleus of M31 has only a $\lesssim$0.1% chance of experiencing a strong dynamical encounter with any type of single star over a time of 10 Gyr (eq. [4] with $s/R_\odot \lesssim 100$ and $I = 1$). A binary that does have a successful exchange encounter with a NS will most likely never experience a second perturbative interaction, unless its orbit takes it to $r \ll 10\,$pc where the density is much higher.

Finally, we summarize the total volumetric rates of collision and exchange in the region $r =$ 10–100 pc, in which case $\xi_{\rm min}^{-1} - \xi_{\rm max}^{-1} \simeq 1$. Using appropriate scalings for the free parameters, including a binary fraction of 10%, and the numerical results given above, we find

Direct Collision:

$$\Gamma_c(10\text{–}100\,{\rm pc}) \;\sim\; 10^{-9}\left(\frac{r_0}{10\,{\rm pc}}\right)^3\left(\frac{n_0}{10^3\,{\rm pc}^{-3}}\right)^2\left(\frac{\beta_t}{1}\frac{\beta_{\rm ns}}{0.01}\frac{M_t}{2\,M_\odot}\frac{s}{R_\odot}\frac{150\,{\rm km\,s^{-1}}}{\sigma}\right)\left(\frac{I_c}{1}\right){\rm yr}^{-1} , \tag{13}$$

Exchange:

$$\Gamma_e(10\text{–}100\,{\rm pc}) \;\sim\; 3\times 10^{-10}\left(\frac{r_0}{10\,{\rm pc}}\right)^3\left(\frac{n_0}{10^3\,{\rm pc}^{-3}}\right)^2\left(\frac{\beta_t}{0.1}\frac{\beta_{\rm ns}}{0.01}\frac{M_t}{3\,M_\odot}\frac{150\,{\rm km\,s^{-1}}}{\sigma}\right){\rm yr}^{-1} . \tag{14}$$

If the dynamical formation rate of LMXBs in this region is $\sim 10^{-9}\,{\rm yr}^{-1}$, then the X-ray lifetime for $L_{\rm X} > 10^{36}\,{\rm ergs\,s^{-1}}$ would have to be $\gtrsim 10^{10}\,$yr in order to observe $\sim$10 sources. A much more reasonable characteristic X-ray lifetime is $\sim 10^8$–$10^9\,$yr, depending on the type of binary. The formation rate could be made $\sim$10 times higher without any trouble, by increasing s and/or n_0 by a factor of $\sim$3. For a rate of $10^{-8}\,{\rm yr}^{-1}$ and a plausible X-ray lifetime of $10^9\,$yr, one should expect to observe $\sim$10 dynamically formed sources.

Appendix.

The total stellar density in the nucleus of M31 may be estimated using the following exercise. Given the surface brightness μ_λ (magnitudes per square arcsecond) in some band, distance $D \simeq 780\,\mathrm{kpc}$ and distance modulus DM $\simeq 24.5$ for M31, and absolute magnitude $M_{\odot,\lambda}$ of the Sun, the intensity in units of $L_\odot\,\mathrm{pc}^{-2}$ is

$$I_\lambda(r) = 10^{-0.4[\mu_\lambda(r)-\mathrm{DM}-M_{\odot,\lambda}]}\left[\frac{206265}{D(\mathrm{pc})}\right]^2 L_\odot\,\mathrm{pc}^{-2}\ . \tag{15}$$

The surface number density of stars equals $\Upsilon_\lambda I_\lambda/\langle m\rangle$, where Υ_λ is the mass-to-light ratio, and $\langle m\rangle$ is the mean stellar mass. If we assume that $n(r) \propto r^{-2}$, which is not far from the truth, then $I_\lambda(r)$ is related to $n(r)$ by

$$n(r) = \frac{\Upsilon_\lambda I_\lambda(r)}{\langle m\rangle \pi r}\ . \tag{16}$$

Kormendy (1988; ApJ 325, 128) shows a B-band surface brightness profile for the center of M31. At an angular radius of $3''$ ($r \simeq 10\,\mathrm{pc}$), his plot gives $\mu_B \simeq 17$, from which we find (with $M_{\odot,B} = 5.5$) $I_B(10\,\mathrm{pc}) \simeq 10^4\,L_\odot\,\mathrm{pc}^{-2}$. Using plausible values of $\Upsilon_B = 3$ and $\langle m\rangle = 0.5\,M_\odot$, we estimate $n(10\,\mathrm{pc}) \simeq 2\times 10^3\,\mathrm{pc}^{-3}$.